Motivation Letter – Zhuoyang Jiang

Application for PhD in the ELLIS Network

I. Research Trajectory: Committed to Building AI for Human Habitats

1. From Passion: Rapidly Applying AI to Urban Planning

My academic journey is rooted in a long-standing fascination with architecture and the design of human living environments. Although I enrolled at ***Southeast University***—one of China's top architecture institutions—my major was reassigned to **Artificial Intelligence** due to limited enrollment quotas in the national college entrance examination. Rather than abandoning my architectural passion, I chose to pursue a new path: from the very beginning of my studies in AI, I envisioned leveraging AI to empower the future of habitat design.

This motivation shaped my undergraduate trajectory. Driven by my unfulfilled architectural aspirations, I proactively collaborated with faculty from the School of Architecture to explore **AI-assisted Solutions to Classical Urban-design Problems**. I constructed a domain-specific dataset and designed an end-to-end intelligent pipeline to modernize a traditional site-selection task. This interdisciplinary project yielded two national patents and a **JCR Q1 journal publication** [1], with its visual featured on the journal's cover. This experience marked my first attempt to use AI as a tool to understand and reshape the built environment.

2. After Reflection: Methodologically Deepening My AI Foundations around Core Pain Points

Despite the early success, I realized the limitations of my AI skill set. The prior project was developed entirely within the architectural domain, lacking methodological guidance from AI experts. I also came to appreciate the decisive role of data—its structure, representation, and task alignment—in determining downstream model effectiveness.

These realizations led me to pursue a Master's in **Data-Centric AI** at ***HKUST(GZ)***, with a clear goal: to build a more solid foundation in AI methodologies that will benefit solving problems of AI for habitat design. During this period, I interned at ***Microsoft Research AI4Science***, where I focused on **Graph Representation Learning for Autoregressive Foundation Models in science**. My research on molecular-graph tokenization enhanced the topological representation capabilities of autoregressive foundation models and improved their understanding of pharmacophores, thereby supporting downstream tasks for property prediction in drug design. The related results **are being prepared for submission to a CORE-A+ Conference and then a Nature Sub-journal**. These experiences deepened my algorithmic intuition and engineering capabilities in large-scale model development, while reinforcing **My Core Methodological Interest: Representation Learning for Multimodal, Structured Domain Data.**

In parallel, recognizing that human experience is integral to intelligent habitat design, I also explored user-oriented research. I participated in several **Human-AI Interaction** projects, including the development of multi-agent systems and the design of interactive interfaces at both HKUST(GZ) and Microsoft Research. Though peripheral to my core research, these efforts deepened my understanding of user interactivity, which I plan to integrate into design-assistive AI systems in the future.

3. Toward Integration: Decisively Resuming Substantive, Genuine Interdisciplinary Research

Now equipped with stronger methodological foundations, I am returning to my original goal: AI for future habitats. My current work for the coming year includes collaborating with architectural researchers on a **Diffusion-based Floorplan Generation Model Constrained by Spatial Syntax at the Data Level**, and conducting research in **Representation Learning for Spatio-temporal Foundation Models in Urban Computing** at the ***CityMind Lab*** at HKUST(GZ). These efforts mark the formal re-entry point of my research into the intersection of AI methods and human habitat design.

II. Research Proposal: Find the Essentials, Focus on Representation Learning

Representation Learning for Design-Assistive Foundation Models on Multimodal, Cross-Scale Habitat Data

Building on the incremental development described above, my PhD will formally focus on representation learning over multimodal data at both the macro urban scale and the human-level building scale, with the goal of enabling design-assistive foundation models for future habitats.

1. Vision: Not Only AI for Habitat, But Also Habitat for AI

Working at the intersection of habitat design and AI, I expand the shared frontier by bringing each field's boundary closer to the other's core. This drives two complementary aims: Not only AI for Habitat, but also Habitat for AI.

- **Not only AI for Habitat:** Use AI to mine and understand patterns across the architecture–city continuum, so as to assist design analysis and design generation in real, verifiable problems about future intelligent habitats.
- **But also Habitat for AI:** The habitat-design domain is not merely an AI application scenario—it is a richly embodied world full of multimodal physical data and human-centered, cross-scale tasks. As such, it functions as a natural playground and wind tunnel for advanced AI, enabling reflection on and feedback into world-model and spatial-intelligence research. By exposing AI with real habitats' spatial topology, temporal dynamics and human-agent interactions, we push models beyond static datasets toward agents that truly perceive, reason and act in the built environment. The methodologies developed here will not only enhance habitat-design applications but also contribute to general artificial intelligence [2, 3, 4].

2. Background: Accumulated Data, Scale-Siloed Models, Weak Cross-Scale Representations

As the built environment becomes increasingly digitized, large-scale, cross-scale multimodal datasets—from Global Earth Observation [5], Urban [6], Residential Floorplan [7] to 3D Indoor Scenes [8]—are rapidly accumulating, laying the groundwork for design-assistive foundation models for human habitats.

Against this backdrop, however, current work [9, 10] on intelligent habitat design remains siloed: "city is city, building is building." Urban foundation models [11, 10] and building-design foundation models [7] are still confined to isolation-scale task settings. They do not yet organize the full data ecosystem of the whole built environment to support truly general foundation models. Achieving this requires a primary emphasis on data and, in particular, on how data are represented.

3. Theme: Cross-Scale, Multimodal Representation Learning for Design-Assistive FMs

From the AI perspective:

- **Core thrust (the entry point and top priority):** Build representation learning methods tailored to future habitat corpora. These corpora are inherently cross-scale (macro urban; human-perceived building) and multimodal (text, graph, image, 3D). We will design data infrastructure and representation methods that are native to the foundation-model era.
- **Backbone and vehicle (the modeling target):** Construct design-assistive foundation models that are general across planning, landscape, and architecture, and that support cross-scale design (for example, using urban-scale signals to guide building design). The key again lies in well-formed data and representations that can scale and generalize.
- **Interaction (the final layer for application readiness):** Once the foundation is mature, enhance human interaction by developing advanced interfaces and agent-based systems to improve the design workflow.

From the habitat-design perspective

- **Within-scale design problems:** floor-plan composition, façade/envelope generation, and spatial-layout reasoning inside buildings.
- **Cross-scale design problems:** using urban-level information (for example, mobility, landscape, or space-syntax indicators) to guide local landscape and architectural design.

4. Plan: Better Single-Scale Representation for FMs → Cross-Scale Fusion → Deployment

Phase 1: Optimize representation learning for single scales (while covering multiple modalities), primarily with autoregressive and diffusion models.

- Building-level: Design briefs (text), Floor plans and elevations (graph or image), 3D models (3D), and so on. [9]
- Urban-level: Points of Interest (graph), Satellite imagery and street-view imagery (image), Urban opinions online (text), Urban mobility trajectories (spatiotemporal), and so on. [6]

Phase 2: Towards a general design-assistive foundation model, define cross-scale tasks and explore representation learning for cross-scale data fusion.

- In collaboration with domain experts, define core cross-scale design problems and provide precise mathematical formulations along the **Earth → City → District → Building → Individual** pipeline.
- Build benchmarks for these tasks and study representation learning for cross-scale fusion.
- Center the effort on dataset processing and construction to conduct large-scale foundation-model training.

Phase 3: Improve interaction and deliver usable design-assistive tools.

Leverage advances in human–AI interaction and agent technology to develop production-quality websites, apps, and plugins for both professional designers and lay users.

5. Standards: Four Directions of Innovation and Three-Axis Evaluation of Solid Work

- For each subproject (corresponding to a conference paper), there must be depth in more than one of the Four Directions of Innovation: **Data-centric**, **Algorithm-driven**, **Domain-based**, or **User-oriented**.
- For each complete phase (corresponding to a journal paper or a usable product), the work must undergo a Three-Axis Evaluation of Solid Work: **Reliability of AI**, **Usability for Human**, and **Beneficial-impact for Habitats**.

III. Program Fit: Bridge AI and Future Habitats at ELLIS

ELLIS offers a unique environment that aligns exceptionally well with my research background and ambitions:

- **AI–Domain Synergy**: ELLIS emphasizes real-world impact while cultivating excellence in fundamental AI research. Its structure—linking advisors focused on methodological AI and advisors focused on application domains—is ideally aligned with my interdisciplinary agenda. As a pan-European network of AI excellence, ELLIS also connects leading research units across geoscience, architecture, design and AI, offering the diverse context I seek.
- **Open and Ethical Research Culture**: shared milestones, open data practices, and interdisciplinary values align with my vision of collaborative, responsible AI for the future habitats.
- **European Model of Scientific Research Training**: I am drawn to the European PhD model, which offers a calm and steady research environment and emphasis on researcher autonomy and their work's lifelong continuity.

My academic path reflects a consistent arc: from architectural dreams to AI foundations, and finally focus on AI for future habitats. I now seek to formalize and scale this integration through doctoral research. I am eager to contribute to the ELLIS network by developing AI that is not only methodologically sound but also meaningfully embedded in the design of our future living environments.

References

[1] Conghui Zhou, Liuyi Yang, Zhuoyang Jiang, and Xinyu Wu. A deep learning approach to identify potential sites for pocket park installation in nanjing, china. *Journal of Urban Technology*, 31(4-5):191–217, 2024.

[2] Xinqing Li, Xin He, Le Zhang, and Yun Liu. A comprehensive survey on world models for embodied ai. *arXiv preprint arXiv:2510.16732*, 2025.

[3] Pascale Fung, Yoram Bachrach, Asli Celikyilmaz, Kamalika Chaudhuri, Delong Chen, Willy Chung, Emmanuel Dupoux, Hongyu Gong, Hervé Jégou, Alessandro Lazaric, et al. Embodied ai agents: Modeling the world. *arXiv preprint arXiv:2506.22355*, 2025.

[4] Xavier Puig, Eric Undersander, Andrew Szot, Mikael Dallaire Cote, Tsung-Yen Yang, Ruslan Partsey, Ruta Desai, Alexander William Clegg, Michal Hlavac, So Yeon Min, et al. Habitat 3.0: A co-habitat for humans, avatars and robots. *arXiv preprint arXiv:2310.13724*, 2023.

[5] Christopher F Brown, Michal R Kazmierski, Valerie J Pasquarella, William J Rucklidge, Masha Samsikova, Chenhui Zhang, Evan Shelhamer, Estefania Lahera, Olivia Wiles, Simon Ilyushchenko, et al. Alphaearth foundations: An embedding field model for accurate and efficient global mapping from sparse label data. *arXiv preprint arXiv:2507.22291*, 2025.

[6] Xingchen Zou, Yibo Yan, Xixuan Hao, Yuehong Hu, Haomin Wen, Erdong Liu, Junbo Zhang, Yong Li, Tianrui Li, Yu Zheng, et al. Deep learning for cross-domain data fusion in urban computing: Taxonomy, advances, and outlook. *Information Fusion*, 113:102606, 2025.

[7] Jun Yin, Pengyu Zeng, Haoyuan Sun, Yuqin Dai, Han Zheng, Miao Zhang, Yachao Zhang, and Shuai Lu. Floorplanllama: Aligning architects' feedback and domain knowledge in architectural floor plan generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6640–6662, 2025.

[8] Qi Sun, Hang Zhou, Wengang Zhou, Li Li, and Houqiang Li. Forest2seq: Revitalizing order prior for sequential indoor scene synthesis. In *European Conference on Computer Vision*, pages 251–268. Springer, 2024.

[9] Jiachen Liu, Yuan Xue, Haomiao Ni, Rui Yu, Zihan Zhou, and Sharon X Huang. Computer-aided layout generation for building design: A review. *arXiv preprint arXiv:2504.09694*, 2025.

[10] Yu Zheng, Fengli Xu, Yuming Lin, Paolo Santi, Carlo Ratti, Qi R Wang, and Yong Li. Urban planning in the era of large language models. *Nature computational science*, pages 1–10, 2025.

[11] Jie Feng, Shengyuan Wang, Tianhui Liu, Yanxin Xi, and Yong Li. Urbanllava: A multi-modal large language model for urban intelligence. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 6209–6219, 2025.